



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
12-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A planet revolves around the sun in an elliptical orbit of eccentricity e . If T is the time period of the planet, then the time spent by the planet between the ends of the minor axis close to the sun, is :-

(A) $T \left[\frac{1}{4} - \frac{e}{2\pi} \right]$

(B) $\frac{T \times e}{\pi}$

(C) $T - \left[\frac{e}{\pi} + \frac{1}{2} \right]$

(D) $T \left[\frac{1}{2} - \frac{e}{\pi} \right]$

2. A conducting sphere S_1 of radius r is attached to an insulating handle. Another conducting sphere S_2 of radius R is mounted on an insulating stand, S_2 is initially uncharged. S_1 is given a charge Q . It is brought into contact with S_2 and removed. S_1 is recharged such that the charge on it is again Q and it is again brought into contact with S_2 and removed. This procedure is repeated n times. Then, the charge on S_2 after n such contacts is :-

(A) $Q \left(\frac{R}{r} \right)^n$

(B) $Q \left(\frac{r}{R} \right)^n$

(C) $\frac{QR}{r} \left[1 - \left(\frac{R}{R+r} \right)^n \right]$

(D) $\frac{QR}{n} \left[1 - \left(\frac{R+r}{R} \right)^n \right]$

Space for Rough Work

3. A particle moves in a straight line starting from rest with constant acceleration and its distances from the origin O on the line at times t_1 , t_2 and t_3 are x_1 , x_2 and x_3 respectively. If t_1 , t_2 , t_3 form an arithmetic progression whose common difference is d and x_1 , x_2 , x_3 are in geometric progression, then the acceleration is:

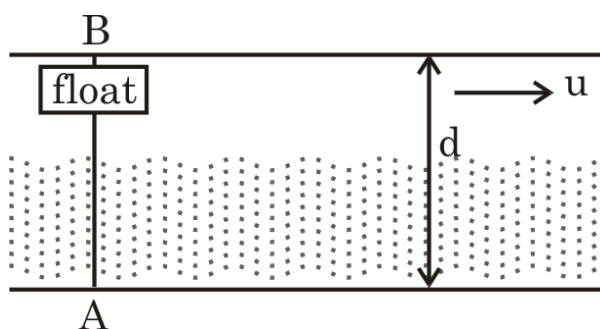
(A) $\frac{\sqrt{x_1} + \sqrt{x_2}}{d^2}$

(B) $\frac{(\sqrt{x_1} - \sqrt{x_3})^2}{d^2}$

(C) $\frac{x_1 - x_2}{\sqrt{d}}$

(D) $\frac{(\sqrt{x_1} - \sqrt{x_3})^2}{2d^2}$

4. A river is flowing with a speed of u and a float is flowing with the river. At an instant when float crosses the line AB near point B. A man jumps in the river from point A to catch the float in minimum time. Then choose the correct option :



- (A) He should swim perpendicular to river flow.
 (B) He should swim down the stream.
 (C) If velocity of river is greater than velocity of swimmer w.r.t. river, he can't catch the float.
 (D) None of these.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

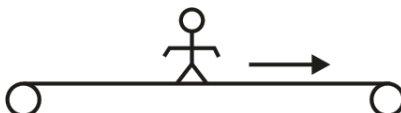
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

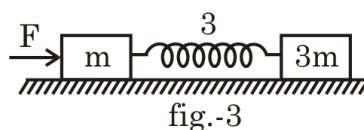
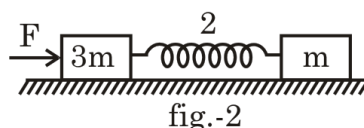
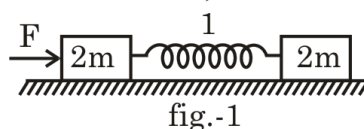
5. The figure below shows a man standing stationary with respect to a horizontal conveyor belt which is accelerating at 1 m/s^2 . Take mass of person as 70 kg. If the maximum acceleration of the belt for which the man remains stationary with respect to belt is 3 m/s^2 then



- (A) The coefficient of friction between man's shoes and the belt is 0.2.
- (B) The coefficient of friction between man's shoes and the belt is 0.3.
- (C) Net force on the man is zero.
- (D) Net force on the man is 70 N.

Space for Rough Work

6. A ring of radius R has charges q and $2q$ distributed uniformly on its two halves. There is a point P on its axis at a distance of $2\sqrt{2}R$ from its center. Which of the following options about the electric field and potential at P is/are correct ?
- (A) Potential is Kq/R
- (B) Component of field along the axis at P is $\frac{2\sqrt{2}Kq}{9R^2}$
- (C) Component of field perpendicular to axis at P is $\frac{2Kq}{9R^2}$
- (D) Component of field perpendicular to axis at P is $\frac{2Kq}{27\pi R^2}$
7. In each of three figures shown, two blocks are connected by a light spring and the system is placed on smooth horizontal surface. A constant horizontal force of magnitude F is applied to left block as shown. Assuming spring constant in all three cases to be same, which of the following statements is/are true.



- (A) Maximum compression in spring 1 is greater than that in spring 2.
- (B) Maximum compression in spring 3 is greater than that in spring 1.
- (C) Maximum compression in spring 3 is greater than that in spring 2.
- (D) Maximum compression in all springs is equal.

Space for Rough Work

8. The potential energy of a particle has a form $U = a \left(\frac{x}{y} - \frac{y}{z} \right)$ where a is constant.

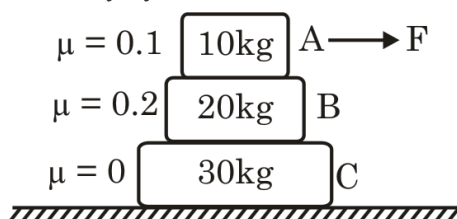
(A) The magnitude of force acting on the particle at $(1, 1, 1)$ is $\sqrt{6}a$ N

(B) The magnitude of force acting on the particle at $(2, 3, 5)$ is $\sqrt{24}a$ N

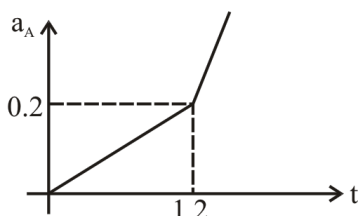
(C) The work done by force in displacing the particle from $(1, 1, 1)$ to $(2, 3, 4)$ is $\frac{a}{12}$ J

(D) The work done by force in displacing the particle from $(1, 1, 1)$ to $(2, 3, 4)$ is $-\frac{a}{12}$ J

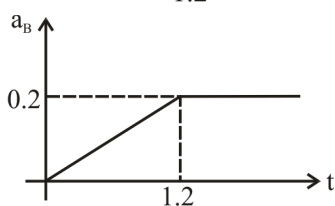
9. A block A is placed on B and B is placed on block C, which rests on a smooth horizontal ground as shown in the figure. Block A is pulled horizontally by a force F which increases gradually. If $F = (10t)$ N,



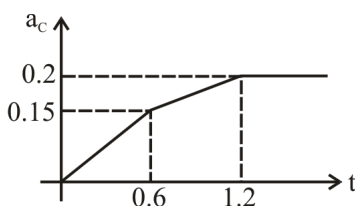
- (A) The acceleration of block A changes with time as



- (B) The acceleration of block B changes with time as



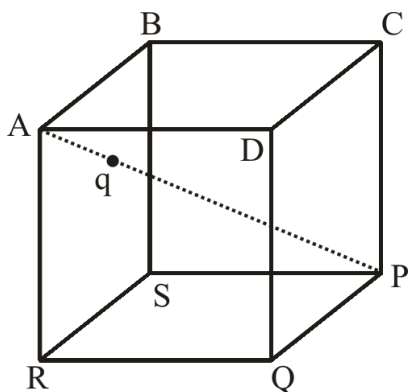
- (C) The acceleration of block C changes with time as



- (D) None of the above

Space for Rough Work

10. A charge q is placed on the diagonal AP of a cube at a distance $AP/3$ from the point A . Choose the correct options.



- (A) the sum of electric flux passing through the surfaces $ABCD$ and $PQRS$ is $q/3\epsilon_0$
 (B) the sum of electric flux passing through the surfaces $ABCD$ and $PQRS$ is $q/8\epsilon_0$
 (C) the flux through both the surfaces $ABCD$ and $PQRS$ are same
 (D) the flux through the surfaces $ABCD$ is larger than the flux through surface $PQRS$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In a certain region, there are non-uniform electrical potential (V_e) as well as gravitational potential (V_g). The electrical potential varies only with x as shown in figure (i) and the gravitational potential varies only with y as shown in figure (ii). Consider a particle of mass 200 kg and charge $20\mu\text{C}$ in this field.

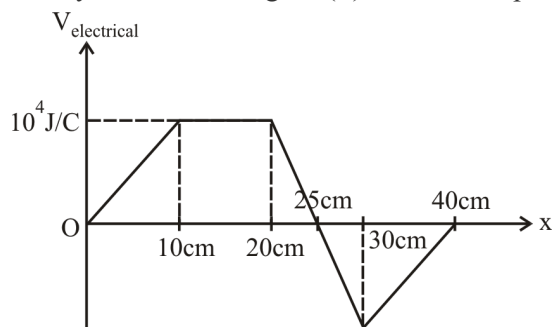


Figure-i

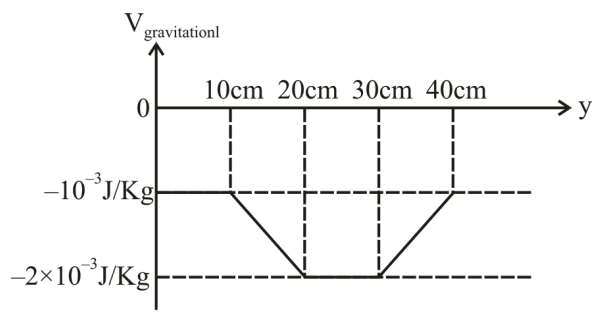


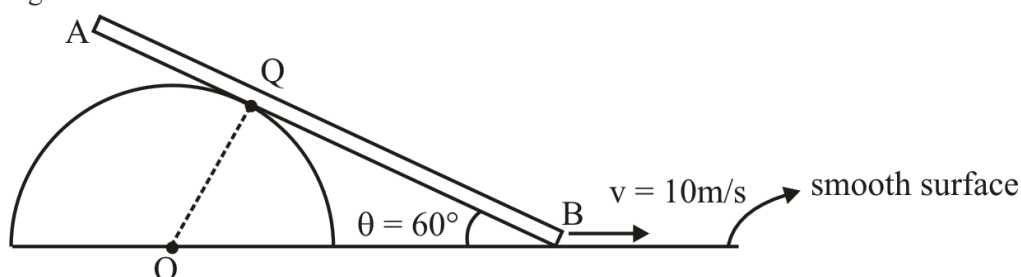
Figure-ii

1. The magnitude of acceleration of the particle at point (25, 35) cm (in cm/sec^2) is
2. Work done (in J) by the external agent to slowly bring the particle from (5, 15) cm to (25, 35) cm is

Space for Rough Work

Paragraph for Questions 3 and 4

A slender rod AB rests on fixed smooth hemisphere of radius $R = 10$ m. End 'B' moves horizontally at speed 10 m/s. Q is a point on the rod which is in contact with the hemisphere at this instant. Then answer the following :

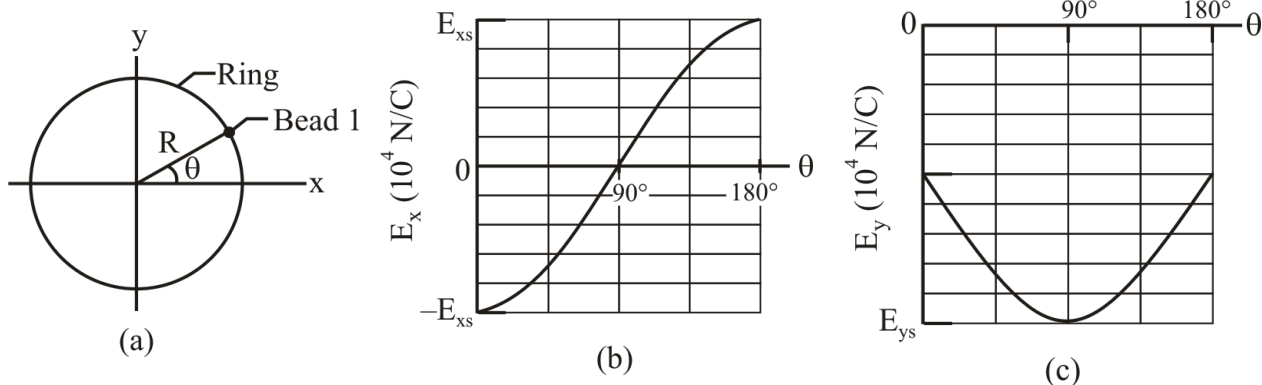


3. The speed of point Q (in m/s) is
4. Angular speed of rod at given instant (in rad/sec) is

Space for Rough Work

Paragraph for Questions 5 and 6

Two charged beads are on the plastic ring in figure a. Bead 2, which is not shown, is fixed in place on the ring, which has radius $R = 60.0$ cm. Bead 1 is initially on the x axis at angle $\theta = 0^\circ$. It is then moved to the opposite side, at angle $\theta = 180^\circ$, through the first and second quadrants of the xy coordinate system. Figure b gives the x component of the net electric field produced at the origin by the two beads as a function of θ , and figure c gives the y component. $E_{xs} = 5.0 \times 10^4$ N/C and $E_{ys} = -9.0 \times 10^4$ N/C.



5. The angle θ (in degree) with the x-axis in the counter clockwise sense at which bead 2 is located is
6. The ratio of charges on bead 1 and bead 2 is

Space for Rough Work

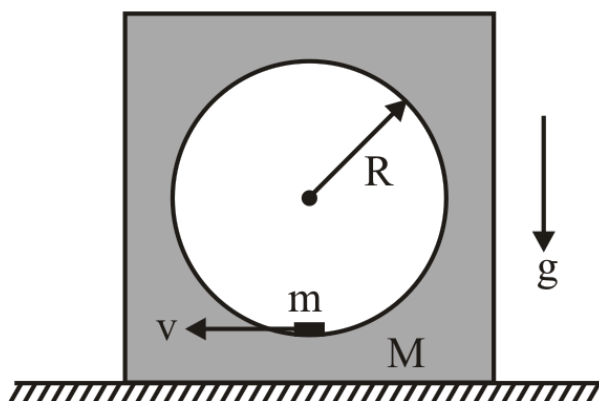
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

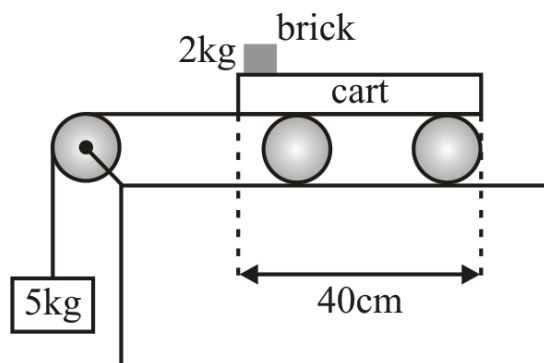
Zero Marks : 0 In all other cases

7. The body is a cube in the center of which a spherical cavity of radius R is carved out. Inside the spherical cavity at its bottom there is a motionless puck whose geometric sizes are negligible. The minimal horizontal velocity (at all possible cube-to-puck mass ratios) has to be imposed on the puck so as the cubic body should jump up from the table surface in the motion process followed. Friction in the system is completely absent. At which value of the cube-to-puck mass ratio M/m that minimal velocity is achieved?

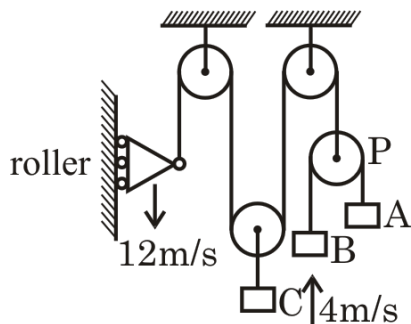


Space for Rough Work

8. A cart of mass 3 kg is pulled by a 5 kg object as shown in figure. Length of cart is 40 cm and the table below it is frictionless. There is a small brick of mass 2 kg on the cart which falls at 0.8 sec after the start of motion. If the coefficient of friction between the cart and the brick is x . The value of $10x$ is. (Take $g = 10 \text{ m/s}^2$)



9. In the arrangement shown in figure, at a particular instant, the roller is coming down with a speed of 12 m/s and C is moving up with 4 m/s. At the same instant, it is also known the wrt pulley P, block A is moving down with speed 3 m/s. Determine the speed of block B (velocity) wrt ground (in m/s).



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

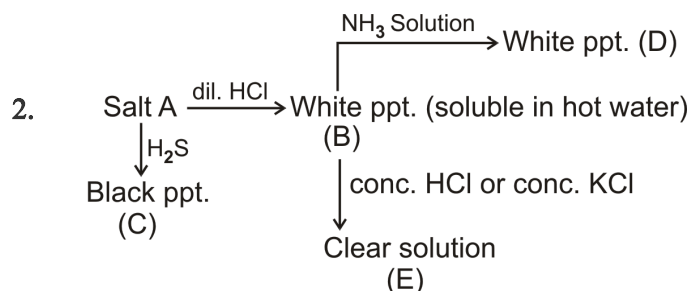
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Choose incorrect statement.

- (A) Tritium, a radioactive isotope of hydrogen, emits beta (β^-) particles
- (B) In basic medium H_2O_2 act as a reducing agent with Fe^{3+} ion
- (C) Polyphosphates are used as water softening agents because they form precipitate with cationic species in hard water
- (D) When calcium carbide react with heavy water CD_4 is formed

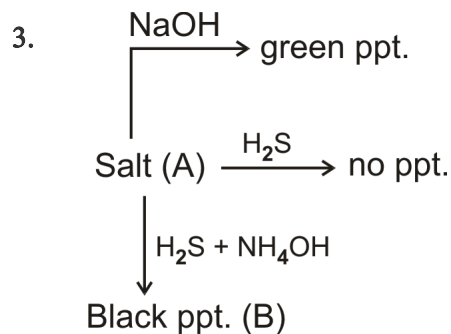


Consider aqueous solution of Salt A.

Choose correct statement.

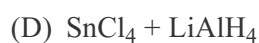
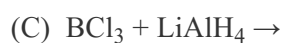
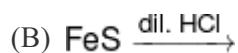
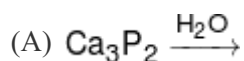
- (A) White ppt D is soluble in excess NaOH
- (B) ppt (B) and ppt (D) has same molecular formula
- (C) Black ppt. (C) soluble in H_2O_2 solution
- (D) Formula of E is $[\text{PbCl}_6]^{4-}$

Space for Rough Work



Choose correct statement.

- (A) Salt (A) give brown ppt. with 1-nitroso-2-naphthol reagent
 (B) Salt (A) give brown ppt. with DMG^{-1}
 (C) Black ppt. (B) soluble in dil. CH_3COOH
 (D) Black ppt. (B) soluble in dil. HCl
4. In which of the following reaction electron deficient hydride is formed.



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Choose correct statement/s.

(A) Acidic strength : $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O} < \text{HF}$

(B) The superoxide MO_2 contains the paramagnetic ion O_2^- which is stable only in the presence of large cation such as K, Rb, Cs

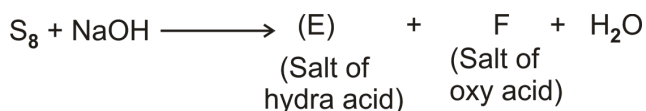
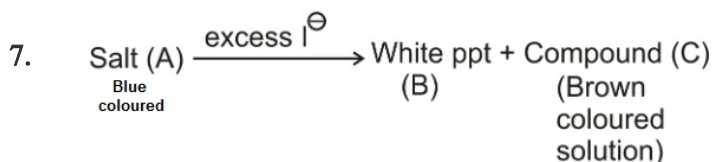
(C) Dielectric constant : $\text{H}_2\text{O} > \text{D}_2\text{O}$

(D) Heat of ionization : $\text{H}_2\text{O} < \text{D}_2\text{O}$

Space for Rough Work

6. Choose correct statement/s.

- (A) Ce^{4+} is a very strong oxidizing agent in aq. solution and is used as a volumetric standard in redox titrations.
- (B) Absorption spectra of the of the coloured lanthanide cations are typically sharp. [absorption bands are narrow]
- (C) Eu and Yb have largest atomic radii among lanthanoids
- (D) Eu^{3+} and Yb^{3+} have largest ionic radii among lanthanoid ions



Choose correct statement.

- (A) Salt (A) + aq. solution of (E) give black ppt.
- (B) Brown colour solution + aq. solution of F give colourless solution
- (C) Aq. solution of (E) can distinguish between Sn^{2+} and Sn^{4+} ion
- (D) Salt F is used in black and white photography

8. Choose correct statement.

- (A) NO_2 is a lung irritant that can lead to an acute respiratory disease in children
- (B) Photochemical smog has ozone, nitric oxide, acrolein, formaldehyde and peroxy acetyl nitrate (PAN) & it is reducing smog
- (C) Due to ozone hole melanoma is produce in human being
- (D) Maximum upper limit of Cd is 0.05 ppm. in drinking water

Space for Rough Work

9. Choose correct statement/s.

- (A) Clean water would have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more
- (B) The F^- ion make the enamel on teeth much harder by converting hydroxyapatite $[Ca_3(PO_4)_2 \cdot Ca(OH)_2]$, the enamel of the surface of the teeth, into much harder fluorapatite $[3Ca_3(PO_4)_2 \cdot CaF_2]$
- (C) Powder of kernel of tamarind seed has been found to be an effective material to make municipal and industrial waste water clean
- (D) Green house gases are CO_2 , Methane, water vapour, nitrous oxide, CFCs and ozone.

10. Choose correct statement/s.

- (A) Melting point order : $Cr > Fe > Mn$
- (B) Colour : (V^{2+} : violet) , (V^{3+} : green) , (V^{4+} : blue)
- (C) $KClO_4$ is isostructural with $KMnO_4$ but $KClO_4$ is white while $KMnO_4$ is purple
- (D) All 3d metal have -ve electrode potential (E°/V) for M^{2+}/M

Space for Rough Work

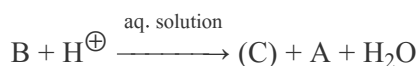
SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

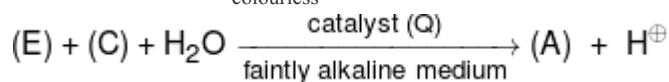
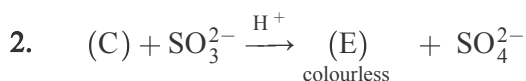
Paragraph for Questions 1 and 2



Compound (A) give greenish yellow gas & compound (D) with HCl

Compound (D) give violet bead in oxidizing flame in borax bead test.

1. If total number of π -bond in (C) is x & total number of $109^\circ 28'$ angle is y than value of (x + y).



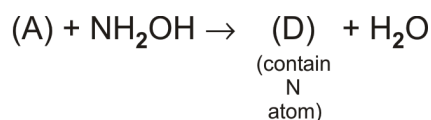
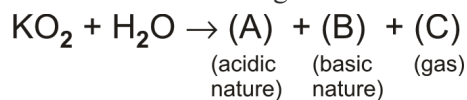
Write total number of d-electron/s present in cationic part of catalyst (Q)

Space for Rough Work

Paragraph for Questions 3 and 4

The increasing stability of the peroxide and superoxide as the size of the alkali ion increases is noteworthy and is a typical example of the stabilization of larger anion by larger cation through lattice energy effects.

3. Consider the following reaction

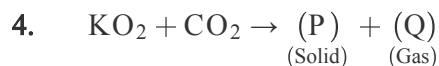


if x = Total number of unpaired electron in gas (C)

y = Maximum number of atom in a plane of compound (A)

z = Maximum number of atom in a plane of compound (D)

then calculate $\frac{(x+y)}{z}$?



Total number of cations among following which give white ppt. with aqueous solution of (P).

Pb^{2+} , Hg^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} , Cu^{2+} , Fe^{3+} , Cr^{3+}

Space for Rough Work

Paragraph for Questions 5 and 6

Binary compound of the elements with hydrogen are called hydrides. Hydrides in three classes.

1. Ionic hydride or salt like hydrides
2. Covalent or molecular hydride
3. Metallic or interstitial hydride

5. Consider the following/statements.

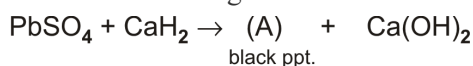
- (a) H_2 has less bond dissociation energy than D_2
- (b) Molten LiH conduct electricity & H_2 liberated at the anode during electrolysis, thus conforming the presence of the hydride ion H^-
- (c) Alkali metal hydride are more reactive than alkaline earth metal hydride
- (d) Except for LiH all I^{st} group hydrides decomposes into their constituent elements on strong heating
- (e) Ionic hydride react with water give H_2
- (f) H_2 molecule exist in two different forms known as ortho and para hydrogen on the basis of nucleus spin

if x = total number of incorrect statement

y = total number of correct statement

then calculate $\frac{x}{y}$.

6. Consider following reaction/s.

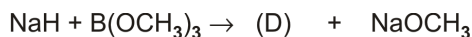


black ppt.



(C)

white ppt.



(E) + NaI + H_2

highly
inflammable

where compound (D) is ionic solid.

if x is maximum number of identical angle in compound (E) and y is maximum number of identical angle in anionic part of white ppt. (C) then calculate value of $x - y$.

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Consider the following order :-

- (a) Ionization energy : $B > Be$
- (b) Electron affinity : $N < Be$
- (c) Electronegativity : $Al > Ga$
- (d) Atomic radius : $Al < Ga$
- (e) Acidic strength : $HClO < HClO_2$
- (f) Hydration energy : $Li^+ < Mg^{2+}$
- (g) Metallic character : $Li > Be$

Calculate value of $(x - y)$ where x and y are incorrect and correct order respectively.

Space for Rough Work

8. (X) is total number of lanthanoids which give H_2 with conc. HCl.

La, Ce, Nd, Pm, Eu, Gd, Th, Pa, U, Np, Am

(Y) is total number of actinoid(s)

Th, Pa, U, Np, Pu, Os, Am, Cm, Ti, Bk, Cf

then calculate value of $(y - x)$.

9. A water sample contain following concentration.

(a) Lead 60 ppb

(b) Sulphate 600 ppm

(c) Nitrate 60 ppm

(d) Fe 0.3 ppm

(e) Mn 0.5 ppm

(f) Al 0.02 ppm

(g) Cu 4.0 ppm

(h) Zn 6.0 ppm

(i) Cd 0.05 ppm

Total number of metal/s or ion/s which are more from upper limit of international standard for drinking water.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If $p(x) = ax^3 + bx^2 + cx + d$ and $q(x) = \ell x^3 + mx^2 + nx + r$ with $p(1) = q(1)$; $p(2) - q(2) = 1$ and $p(3) - q(3) = 4$, then $p(4) - q(4)$ is (Given $a - \ell = 1$)
(A) 0 (B) 9 (C) 6 (D) 15
 2. If $S = \sum_{k=1}^{10} \sum_{r=0}^{10} (10 + k - r)^2$, then the value of $\frac{S}{55}$.
(A) 243 (B) 257 (C) 355 (D) 293
 3. Let a and b are two positive real numbers such that $a + b = 1$. If the minimum value of $\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2$ is, M then the value of $2M$ equals
(A) 50 (B) $\frac{25}{2}$ (C) 25 (D) 100
 4. A circle touches the parabola $y^2 = 8x$ at P . It also passes through the focus S of the parabola and intersects its axis at Q . If angle SPQ is 90° , and equation of the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $(2g + 3f + c)$ equals (Q and S are distinct points)
(A) 16 (B) 20 (C) 24 (D) 12
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Suppose $a, b > 0$ and x_1, x_2, x_3 ($x_1 > x_2 > x_3$) are roots of $\frac{x-b}{a} + \frac{x-a}{b} = \frac{b}{x-a} + \frac{a}{x-b}$ and $x_1 - x_2 - x_3 = c$, then

(A) a, c, b are in H.P. and $x_1 = a + b$

(B) a, c, b are in A.P. and $x_2 = a + b$

(C) a, c, b are in A.P. and $x_3 = 0$

(D) a, c, b are in H.P. and $x_3 = 0$

Space for Rough Work

6. If $(x - a)\cos\theta + y\sin\theta = a = (x - a)\cos\phi + y\sin\phi$ and $\tan\left(\frac{\theta}{2}\right) - \tan\left(\frac{\phi}{2}\right) = 2b$, then
- (A) $y^2 = 2ax - (1 - b^2)x^2$ (B) $\tan\frac{\theta}{2} = \frac{y + bx}{x}$
- (C) $y^2 = 2bx - (1 - a^2)x^2$ (D) $\tan\frac{\phi}{2} = \frac{y - bx}{x}$
7. ℓ, m are variable real numbers such that $5\ell^2 + 6m^2 - 4\ell m + 3\ell = 0$. If variable line $\ell x + my = 1$ always touches a fixed parabola, whose axis is parallel to x axis. Then
- (A) Focus be $\left(\frac{-1}{3}, \frac{-4}{3}\right)$ (B) Equation of directrix is $3y + 11 = 0$
- (C) Focus be $\left(\frac{1}{3}, \frac{4}{3}\right)$ (D) Equation of directrix is $3x + 11 = 0$
8. If three distinct chords drawn from $(\alpha, 0)$ to the ellipse $x^2 + 2y^2 = 1$ are bisected by the parabola $y^2 = 4x$ then α lies in the interval.
- (A) (2, 10) (B) $(8, 4 + \sqrt{17})$ (C) (0, 16) (D) (-7, 8)
9. Circles are drawn with OA & OB as diameters, where A & B are points of parabola $y^2 = 4x$. These circles meet at P (other than O). m_1 and m_2 are slope of tangents to the parabola $y^2 = 4x$ at A & B respectively and m is slope of chord AB, then (given $m_1 + m_2 \neq 0$, A, B are points other than origin and 'O' is origin)
- (A) A, P, B are collinear points (B) m is harmonic mean of m_1 and m_2
- (C) m is arithmetic mean of m_1 and m_2 (D) OP is perpendicular to AB
10. From centre O, of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, two perpendicular rays are drawn meeting the ellipse at P & Q, N is the foot of perpendicular from O to PQ, then
- (A) $\frac{1}{OP^2} + \frac{1}{OQ^2} = \frac{25}{144}$ (B) $\frac{1}{OP^2} - \frac{1}{OQ^2} = \frac{25}{144}$
- (C) $ON = \frac{12}{5}$ (D) $ON = \frac{6}{5}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

$y = ax^2 + bx + c$ is defined for $x \in [0, 2]$ and $y \in [0, 1]$ such that $2ax + b = 0$ has a solution which does not belong to interval $(0, 2)$.

1. Then the value of $|2a + b|$ is
2. In previous question, if sum of roots of $ax^2 + bx + c = 0$ is $(p^2 + p + 3)$ for $p \in \mathbb{R}$, then maximum value of a is

Space for Rough Work

Paragraph for Questions 3 and 4

A point P moves such that sum of the slopes of the normals drawn from it to the hyperbola $xy = 16$ is equal to the sum of the ordinates of the feet of the normals. Let 'P' lies on the curve C, then

3. Length of latus rectum of conic 'C' equals
4. Area of the equilateral triangle, inscribed in the curve C and having one vertex same as the vertex of C is :
(Take $\sqrt{3} = 1.70$)

Paragraph for Questions 5 and 6

In a ΔABC , if $\cos A \cos B \cos C = \frac{\sqrt{3} - 1}{8}$ and $\sin A \sin B \sin C = \frac{3 + \sqrt{3}}{8}$, then

5. The value of $\tan A \tan B + \tan B \tan C + \tan C \tan A$ is :
(Take $\sqrt{3} = 1.70$)
6. If the angles of ΔABC are A, B, C and $\angle A > \angle B > \angle C$ then $\frac{\angle B}{\angle C}$ equals

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If tangents PQ and PR are drawn from a variable point P to the hyperbola $\frac{(x-2)^2}{9} - \frac{(y+1)^2}{4} = 1$ such that fourth vertex S of parallelogram PQRS lies on circumcircle of ΔPQR then locus of P is $x^2 + y^2 + 2gx + 2fy + c = 0$. Find the value of $(g^2 + f^2 + c^2)$
8. Let S denote the sum of the series $\frac{3}{2^3} + \frac{4}{2^4 \cdot 3} + \frac{5}{2^6 \cdot 3} + \frac{6}{2^7 \cdot 5} + \frac{7}{2^7 \cdot 15} + \dots \infty$, then the value of S^{-1} is
9. It is given that for polynomial $p(x) = x^3 + ax^2 + bx + c$, equation $p(x) = 0$ has three distinct positive integral roots α, β, γ . Also $p(14) = 231$. Number of possible unordered triplets of (α, β, γ) is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work